Rates of change and trigonometric derivatives

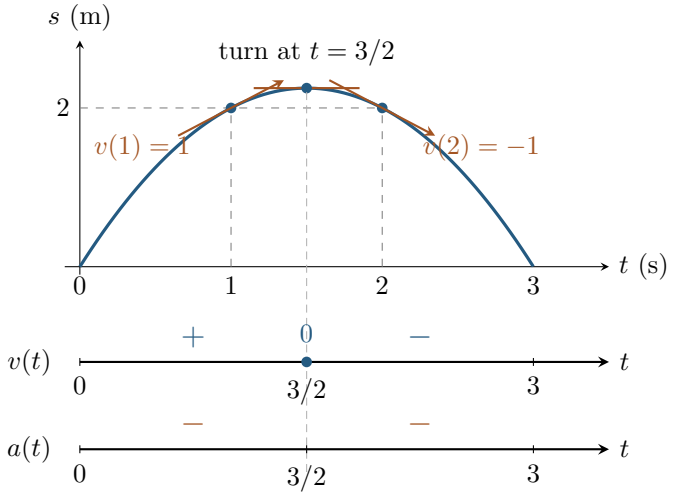
MATH 1334 Friday, October 16, 2026
Sections 3.4 and 3.5

1. From derivatives to motion

Revisit $s(t) = 3t - t^2$, $0 \le t \le 3$, with position in meters and time in seconds. What do its derivatives tell us about the motion?

$$\underbrace{v(t) = s'(t) = 3 - 2t}_{\text{velocity (m/s)}}, \qquad \underbrace{a(t) = v'(t) = -2}_{\text{acceleration (m/s}^2\text{)}}.$$

The sign of velocity gives direction. Speed is $|v(t)|$.



On $(0, 3/2)$, $v > 0$ and $a < 0$: the particle moves in the positive direction and slows down. At $t = 3/2$, it is at rest and reverses direction. On $(3/2, 3)$, $v < 0$ and $a < 0$: it moves in the negative direction and speeds up.

Compare the signs: matching signs of v and a mean speeding up; opposite signs mean slowing down. Negative acceleration alone does not mean slowing down.

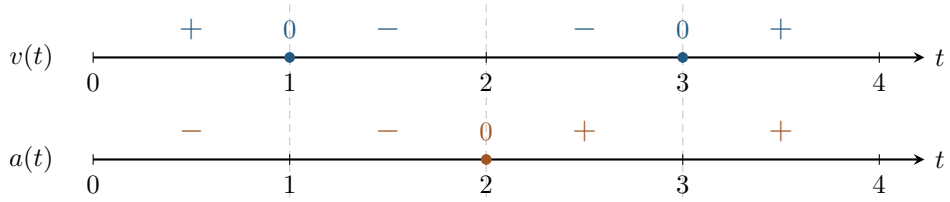
Example 1. A particle has position

$$s(t) = t^3 - 6t^2 + 9t + 2, \qquad 0 \le t \le 4.$$

Find when it is at rest, its direction of motion, and when its speed increases.

$$v(t) = 3t^2 - 12t + 9 = 3(t-1)(t-3), \qquad a(t) = 6t - 12 = 6(t-2).$$

The particle is at rest at $t = 1, 3$. To discuss speed, split the time interval at the zeros of *both* velocity and acceleration:



Use the factored expressions or a test time to find each sign. From the velocity sign graph, motion is positive on $(0, 1)$ and $(3, 4)$ and negative on $(1, 3)$. Comparing the two graphs, the particle speeds up on $(1, 2)$ and $(3, 4)$ and slows down on $(0, 1)$ and $(2, 3)$.

On $(1, 2)$, velocity is negative and becoming more negative, so speed increases. At $t = 2$, acceleration changes sign but velocity does not: speed changes from increasing to decreasing without a change in direction.

2. Read the event before substituting

A question may specify a time, a position, or an event such as impact. The input to a velocity function $v(t)$ is always time. Find that time first.

Example 2. A ball has height $H(t) = 96 - 16t - 16t^2$ feet until it hits the ground, with t measured in seconds. Find the impact time, impact velocity, and impact speed.

Ground level means height zero:

$$96 - 16t - 16t^2 = -16(t+3)(t-2) = 0.$$

The roots are -3 and 2. We reject the negative time, so impact occurs at $t = 2$ s and the flight model applies on $0 \le t \le 2$.

$$v(t) = H'(t) = -16 - 32t, \qquad a(t) = -32.$$

Just before impact, the velocity is $v(2) = -80$ ft/s; the speed is $|v(2)| = 80$ ft/s. The negative sign gives the downward direction. The polynomial can be evaluated after $t = 2$, but those values do not describe this flight after contact with the ground.

Example 3. A car's position is $s(t) = t^2 - 2t - 3$ feet, $t \ge 0$. Find its velocity when (a) $s = 0$ and (b) $s = 5$.

Solve for time before evaluating velocity. First, $v(t) = s'(t) = 2t - 2$ ft/s. Now solve each position condition:

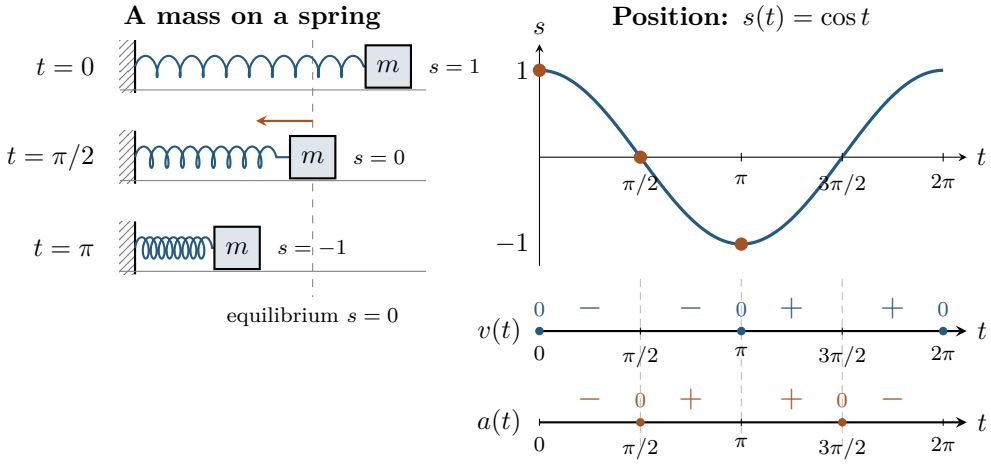
$$\begin{aligned} &\text{(a)} \quad t^2 - 2t - 3 = (t-3)(t+1) = 0 && \Longrightarrow t = 3, \\ & && v(3) = 4 \text{ ft/s}; \\ &\text{(b)} \quad t^2 - 2t - 3 = 5 \Longleftrightarrow (t-4)(t+2) = 0 && \Longrightarrow t = 4, \\ & && v(4) = 6 \text{ ft/s}. \end{aligned}$$

In each case the other root is outside $t \ge 0$. Substituting 5 directly into $v(t)$ would find velocity at five seconds, not at position five feet. In other models, more than one admissible time can satisfy a position condition.

3. Derivatives of trigonometric functions

A mass on a spring has position $s(t) = \cos t$, measured in meters from equilibrium, with t in seconds. Consider one cycle, $0 \le t \le 2\pi$. Using the trigonometric derivatives below,

$$v(t) = s'(t) = -\sin t, \qquad a(t) = v'(t) = -\cos t.$$



The mass moves left on $(0, \pi)$ and right on $(\pi, 2\pi)$. Matching signs give speeding up on $(0, \pi/2)$ and $(\pi, 3\pi/2)$; opposite signs give slowing down on $(\pi/2, \pi)$ and $(3\pi/2, 2\pi)$. It is at rest at $t = 0, \pi, 2\pi$ and moves fastest as it passes equilibrium.

Trigonometric notation we will use

Angles are measured in **radians**; one full turn is 2π radians. On the unit circle, the point at angle x has coordinates $(\cos x, \sin x)$. For a calculator or graphing tool, select radian mode. We use

$$\tan x = \frac{\sin x}{\cos x}, \quad \cot x = \frac{\cos x}{\sin x}, \quad \sec x = \frac{1}{\cos x}, \quad \csc x = \frac{1}{\sin x}.$$

The symbols sec and csc denote reciprocals, not inverse functions. A square such as $\sin^2 x$ means $(\sin x)^2$.

Theorem (Trigonometric derivative formulas)**.** *For angles measured in radians, on the domain of each function,*

$$\frac{d}{dx}(\sin x) = \cos x, \qquad \frac{d}{dx}(\cos x) = -\sin x,$$
$$\frac{d}{dx}(\tan x) = \sec^2 x, \qquad \frac{d}{dx}(\cot x) = -\csc^2 x,$$
$$\frac{d}{dx}(\sec x) = \sec x \tan x, \quad \frac{d}{dx}(\csc x) = -\csc x \cot x.$$

The functions $\tan, \sec$ *require* $\cos x \neq 0$*;* $\cot, \csc$ *require* $\sin x \neq 0$*. Sine and cosine are defined for every real* x*.*

The sine and cosine formulas follow from the limit definition using the angle-addition identities. We take those two formulas as our starting facts here; the quotient rule will explain the other four.

Example 4. Differentiate $G(x) = 3\sin x - x^2$.

$$G'(x) = 3\frac{d}{dx}(\sin x) - \frac{d}{dx}(x^2) = 3\cos x - 2x,$$
$$G''(x) = 3(-\sin x) - 2 = -3\sin x - 2.$$

The difference and scalar-multiple rules still apply. Note the minus sign in $(\cos x)'$.

Choose the rule from the structure

Example 5. Differentiate $y = 2x^2 - \sec x$ and $z = 2\csc x + 3/x$.

$$y' = 4x - \sec x \tan x,$$
$$z' = -2\csc x \cot x - \frac{3}{x^2}.$$

For y, require $\cos x \neq 0$. For z, require $\sin x \neq 0$ and $x \neq 0$. The minus sign in $(\csc x)'$ is part of the formula.

Example 6. Differentiate $y = x^3 \cot x$.

$$\begin{aligned} y' &= \frac{d}{dx}(x^3)\cot x + x^3\frac{d}{dx}(\cot x) \\ &= 3x^2 \cot x + x^3(-\csc^2 x) \\ &= 3x^2 \cot x - x^3 \csc^2 x, \qquad \sin x \neq 0. \end{aligned}$$

The cotangent factor stays in the first term and is differentiated in the second.

Example 7. Differentiate $z = \sin x \tan x$.

$$z' = \cos x \tan x + \sin x \sec^2 x, \qquad \cos x \neq 0.$$

This is a complete answer. The product rule does not require an identity to simplify the result.

Example 8. Differentiate $y = \dfrac{\sec x}{x}$.

$$y' = \frac{x(\sec x \tan x) - (\sec x)(1)}{x^2} = \frac{x \sec x \tan x - \sec x}{x^2}.$$

The original domain requires $x \neq 0$ and $\cos x \neq 0$.

Choose the structure before simplifying

For a sum, differentiate term by term. For a product, write both terms. For a quotient, preserve the numerator subtraction and square the denominator. Identify the structure first, then look up the basic derivative you need.

Using quotients to explain the new formulas

We will use the identity $\sin^2 x + \cos^2 x = 1$.

Example 9. Derive the tangent formula from $\tan x = \sin x / \cos x$.

$$\frac{d}{dx}(\tan x) = \frac{(\cos x)(\cos x) - (\sin x)(-\sin x)}{\cos^2 x}$$
$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x.$$

For cotangent, reverse the roles of sine and cosine:

$$\frac{d}{dx}(\cot x) = \frac{(\sin x)(-\sin x) - (\cos x)(\cos x)}{\sin^2 x} = -\frac{1}{\sin^2 x} = -\csc^2 x.$$

The sign follows from the numerator. These calculations apply on the original domains of tangent and cotangent.

Reciprocals use the same rule. For example,

$$\frac{d}{dx}(\sec x) = \frac{(\cos x)(0) - 1(-\sin x)}{\cos^2 x} = \frac{1}{\cos x}\frac{\sin x}{\cos x} = \sec x \tan x.$$

Example 10. Differentiate $y = \dfrac{\tan x}{2 - \sec x}$.

Keep track of the two minus signs. Since $(2 - \sec x)' = -\sec x \tan x$, the quotient rule gives

$$y' = \frac{(2 - \sec x)\sec^2 x - (\tan x)(-\sec x \tan x)}{(2 - \sec x)^2}$$
$$= \frac{(2 - \sec x)\sec^2 x + \sec x \tan^2 x}{(2 - \sec x)^2}.$$

This form is already correct. To simplify, divide $\sin^2 x + \cos^2 x = 1$ by $\cos^2 x$:

$$\tan^2 x + 1 = \sec^2 x \quad \Longrightarrow \quad \tan^2 x = \sec^2 x - 1.$$

Consequently,

$$(2 - \sec x)\sec^2 x + \sec x \tan^2 x = 2\sec^2 x - \sec^3 x + \sec x(\sec^2 x - 1)$$
$$= 2\sec^2 x - \sec x.$$

Thus $y' = (2\sec^2 x - \sec x)/(2 - \sec x)^2$, still requiring $\cos x \neq 0$ and $\sec x \neq 2$.

Second derivatives and tangent slopes

Example 11. Find y' and y'' if $y = 2\sin x\cos x$.

$$\begin{aligned} y' &= 2\big[(\cos x)(\cos x) + (\sin x)(-\sin x)\big] \\ &= 2\cos^2 x - 2\sin^2 x. \end{aligned}$$

To differentiate again without a new rule, write each square as a product:

$$\begin{aligned} y'' &= 2\big[(-\sin x)\cos x + \cos x(-\sin x)\big] \\ &\quad - 2\big[(\cos x)\sin x + \sin x(\cos x)\big] \\ &= -8\sin x\cos x. \end{aligned}$$

Each product contributes two terms. We have not needed a double-angle formula.

Example 12. Find the tangent to $f(x) = -2\sin x$ at $x = 0$.

Using $\sin 0 = 0$ and $\cos 0 = 1$,

$$f(0) = 0, \qquad f'(x) = -2\cos x, \qquad f'(0) = -2.$$

The tangent line is $y = -2x$. Plot the function and line in radian mode and zoom near $(0, 0)$ to check the result.

Example 13. Find every x in $0 < x < 2\pi$ where $f(x) = x - 4\cos x$ has slope 3.

Turn the slope condition into an equation.

$$f'(x) = 1 + 4\sin x, \qquad 1 + 4\sin x = 3 \quad\Longrightarrow\quad \sin x = \frac{1}{2}.$$

Use these trig facts: sine equals $1/2$ at $\pi/6$ and $5\pi/6$, and these are its only two such inputs between 0 and 2π. Thus $x = \pi/6, 5\pi/6$. Both satisfy the stated interval. The method is: write $f'(x) = m$, solve the resulting equation, and keep every admissible solution.

Practice

State the rule you use, and include units in the applications.

1. A particle has position

$$s(t) = -t^3 + 9t^2 - 24t + 5, \qquad 0 \le t \le 5,$$

with position in meters and time in seconds. Use sign graphs for velocity and acceleration to find when the particle is at rest, its direction of motion, and when it speeds up or slows down.

2. Find y' and y'' if $y = x \sin x$.

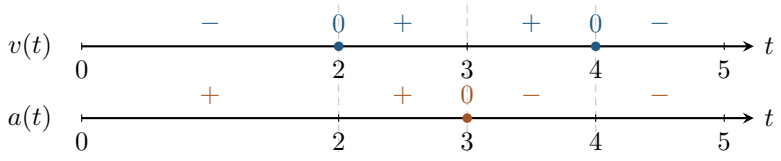
3. Differentiate $y = \dfrac{\sin x}{1 + \cos x}$ and simplify.

Solutions

1. Factor the velocity and mark the zeros of both derivatives:

$$v(t) = -3t^2 + 18t - 24 = -3(t-2)(t-4), \qquad a(t) = 18 - 6t = -6(t-3).$$

Velocity is zero at $t = 2, 4$; acceleration is zero at $t = 3$.



At rest at $t = 2, 4$. Motion is in the positive direction on $(2, 4)$ and in the negative direction on $(0, 2)$ and $(4, 5)$. Matching signs: speeding up on $(2, 3)$ and $(4, 5)$. Opposite signs: slowing down on $(0, 2)$ and $(3, 4)$.

2. Apply the product rule, then differentiate again:

$$y' = \sin x + x \cos x, \qquad y'' = \cos x + \cos x - x \sin x = 2 \cos x - x \sin x.$$

3. Use the quotient rule and $\sin^2 x + \cos^2 x = 1$:

$$\begin{aligned} y' &= \frac{(1 + \cos x)\cos x - (\sin x)(-\sin x)}{(1 + \cos x)^2} \\ &= \frac{\cos x + \cos^2 x + \sin^2 x}{(1 + \cos x)^2} = \frac{1}{1 + \cos x}, \qquad 1 + \cos x \neq 0. \end{aligned}$$